The big reveal: Sharing your findings

Data is great, but if we

can't communicate the story data is telling,

it isn't useful to anyone.

We need ways to organize data that

help us turn it into information.

There are all kinds of tools out there to help you

visualize and share

your data analysis with stakeholders.

Here, we'll talk about

two data presentation tools, reports and dashboards.

Reports and dashboards are both

useful for data visualization.

But there are pros and cons for each of them.

A report is a static collection of

data given to stakeholders periodically.

A dashboard on the other hand,

monitors live, incoming data.

Let's talk about reports first.

Reports are great for giving snapshots of

high level historical data for an organization.

For example, a finance firm's monthly sales.

Reports come with a lot of benefits too.

They can be designed and sent out periodically,

often on a weekly or monthly basis,

as organized and easy to reference information.

They're quick to design and easy to

use as long as you continually maintain them.

Finally, because reports use static data or

data that doesn't change once it's been recorded,

they reflect data that's already been cleaned and sorted.

There are some downsides to keep in mind too.

Reports need regular maintenance

and aren't very visually appealing.

Because they aren't automatic or dynamic,

reports don't show live, evolving data.

For a live reflection of incoming data,

you'll want to design a dashboard.

Dashboards are great for a lot of reasons,

they give your team more access

to information being recorded,

you can interact through data by playing with filters,

and because they're dynamic,

they have long-term value.

If stakeholders need to continually access information,

a dashboard can be more efficient than

having to pull reports over and over,

which is a big time saver for you.

Last but not least,

they're just nice to look at.

But dashboards do have some cons too.

For one thing, they take a lot of time to design and

can actually be less efficient than reports,

if they're not used very often.

If the base table breaks at any point,

they need a lot of maintenance to

get back up and running again.

Dashboards can sometimes overwhelm

people with information too.

If you aren't used to

looking through data on a dashboard,

you might get lost in it.

As a data analyst,

you need to decide the best way to

communicate information to your stakeholders.

For example, what if your stakeholders are

interested in the company's social media engagement?

Would a monthly report that tells them

the number of new followers for their page be useful?

Or a dashboard that monitors live

social media engagement across multiple platforms?

Later on, you'll create

your own reports and

dashboards to practice using these tools.

But for now, I want to show you what

a report and a dashboard might look like.

We'll start by using a tool we're

already familiar with, spreadsheets.

Let's see one way

spreadsheet data could be visualized in a report.

This spreadsheet has a data set with

order details from a wholesale company.

That's a lot of information.

From the headers, we can see

different things recorded here,

like the order date,

the salesperson, the unit price,

and revenue for each transaction recorded.

It's all useful information,

but a little hard to wrap your head around.

We want a report that's easier to read.

Let's say your stakeholders want

a quick look at the revenue by salesperson.

Using the data, you could make them a pivot

table with a graph that shows that information.

A pivot table is

a data summarization tool

that is used in data processing.

Pivot tables are used to summarize, sort, re-organize,

group, count, total,

or average data stored in a database.

It allows its users to transform

columns into rows and rows into columns.

We'll actually learn more about pivot tables later.

But I'll show you one really quick.

We'll select the Data menu and click Pivot table button.

It can pull data from this table.

We can just press

create and it'll pull up a new worksheet.

Over here, it gives us

the pivot table fields we can choose from.

Click select, salesperson and revenue.

Just like that, it made a chart for us.

At this point, you can

play around with how the graph looks,

but the information is all there.

Let's move on to dashboards.

If you need a more dynamic way to

share information with your stakeholders,

dashboards are your friend.

You might create something like this Tableau dashboard.

With interactive graphs that

showcase multiple views of the data.

With this, users can change location, date range,

or any other aspect of the data they're viewing

by clicking through different elements on the dashboard.

Pretty cool, right?

Later in this program,

we'll look into how you can make

your own data visualizations.

We have a lot to learn before we get to that.

But I hope this was an exciting first peek at

the different visualization tools

you'll be using as a data analyst.